

Recovery Exam Statistical Mechanics

2 February 2024

Welcome to the recovery exam of Statistical Mechanics of the GEMIF. The exam covers all the topics we have seen during the course.

Summary of key formulas

$$\sinh^2(x) - \cosh^2(x) = 1$$

Question 1 [1 points]

The number of microstates of a system is given by:

$$\Omega = A \exp \left[\gamma (VU)^{\frac{1}{2}} \right]$$

where γ is a constant, V is the volume, and U is the internal energy.

- (a) Calculate the entropy S of the system.
- (b) Calculate the temperature T of the system.
- (c) Does this system satisfy the equipartition theorem? Why?

Solution Question 1:

- (a) Knowing that $S = k_B \ln \Omega(U, V, N)$ then:

$$S = k_B \ln A + k_B \gamma (UV)^{\frac{1}{2}} \quad (1)$$

- (b) The temperature is determined as $\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N, V}$, then:

$$\frac{1}{T} = k_B \gamma \frac{V^{1/2}}{2U^{1/2}} \implies k_B T = \frac{2}{\gamma} \sqrt{\frac{U}{V}} \quad (2)$$

- (c) The system does not satisfy the equipartition theorem because $U \propto V(k_B T)^2$.
To satisfy the equipartition theorem $U \propto N k_B T$ or $V k_B T$

Question 2: [1.5 points]

Consider a system of non-interacting particles that can be in M energy levels $\epsilon_0 < \epsilon_1 < \dots < \epsilon_M$ in contact with a reservoir at temperature T and chemical potential $\mu < \epsilon_0$.

Compute the grand partition function for:

- (a) Classical particles.
- (b) Bosons.

Solution Question 2:

- (a) Classical Particles: In this case we have no limit in the number of particles that can be in each level, but because classical particles are indistinguishable, we have to take into account the extra permutations:

$$\mathcal{Z} = \sum_N \frac{e^{\beta\mu N}}{N!} \sum_{\{n_i\}} e^{-\beta \sum_i \epsilon_i n_i} \quad (3)$$

Note that by considering the different terms for $N = 1, 2, \dots$ we see that:

$$\begin{aligned} \mathcal{Z} &= e^{\beta\mu} \sum_i e^{-\beta\epsilon_i} + \frac{e^{-\beta\mu 2}}{2!} \left(\sum_{(i,j)} e^{-\beta\epsilon_i - \beta\epsilon_j} + \sum_i e^{-\beta\epsilon_i 2} \right) + \dots \\ \mathcal{Z} &= e^{\beta\mu} Z_1 + \frac{e^{-\beta\mu 2}}{2!} Z_1^2 + \frac{e^{-\beta\mu 3}}{3!} Z_1^3 + \dots \\ \mathcal{Z} &= \sum_N \frac{e^{-\beta\mu N}}{N!} Z_1^N = \exp(e^{\beta\mu} Z_1) \end{aligned} \quad (4)$$

- (b) Bosons: We obtain this result by replicating the procedure done in class:

$$\begin{aligned} \mathcal{Z} &= \sum_{\{n_i\}} e^{-\beta \sum_i (\epsilon_i - \mu) n_i} \\ &= \prod_i \sum_{n_i=0}^{\infty} e^{-\beta(\epsilon_i - \mu) n_i} = \prod_i \frac{1}{1 - e^{-\beta(\epsilon_i - \mu)}} \end{aligned} \quad (5)$$

Question 3: [1.0 points]

Find the approximation of the exact magnetization of the Ising model in an external field h :

$$m = kT \frac{\beta e^{2\beta J} \sinh(\beta h)}{\sqrt{e^{2\beta J} \sinh^2(\beta h) + e^{-2\beta J}}} \quad (6)$$

for high temperatures.

Solution Question 3:

At high temperature $\beta \rightarrow 0$. Then $e^{2\beta J} \rightarrow 1$ and $e^{-2\beta J} \rightarrow 1$. That means that:

$$m \approx kT \frac{\beta \sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + 1}} \quad (7)$$

we know that $\sinh^2(x) - \cosh^2(x) = -1$, then:

$$m \approx kT \frac{\beta \sinh(\beta h)}{\sqrt{\cosh^2(\beta h)}} = kT \beta \frac{\sinh(\beta h)}{\cosh(\beta h)} = \tanh(\beta h) \quad (8)$$

Question 4: [1.5 points]

Consider a magnetic system in the presence of an external magnetic field. The Landau free energy for such a system, where spins contribute to the magnetization in a ferromagnetic material, is given by:

$$F(M, T, H) = \frac{1}{2}A(T - T_c)M^2 + \frac{1}{4}BM^4 - H \cdot M$$

where M is the magnetization of the material, T is the temperature, H is the external magnetic field, and A and B are positive constants. The term $-H \cdot M$ represents the interaction energy between the external field and the magnetization of the material.

- (a) Determine the equation for the equilibrium magnetization M_{eq} as a function of temperature T and the external magnetic field H .
- (b) Discuss the validity of this equation.
- (b) What do you expect from the solutions of this equation?

Solution Question 4:

- (a) Given the Landau free energy function in the presence of an external magnetic field:

$$F(M, T, H) = \frac{1}{2}A(T - T_c)M^2 + \frac{1}{4}BM^4 - H \cdot M \quad (9)$$

To find the equilibrium magnetization M_{eq} , we set the derivative of F with respect to M equal to zero:

$$\frac{\partial F}{\partial M} = A(T - T_c)M + BM^3 - H = 0 \quad (10)$$

Solving for M , we get the equilibrium condition:

$$BM^3 + A(T - T_c)M - H = 0 \quad (11)$$

This is a cubic equation in M whose solution gives the equilibrium magnetization as a function of T and H .

- (b) It is valid when M is small, so that the free energy is reasonably approximated by the expansion around $M = 0$.
- (c) If $h \neq 0$, then the function $f(M)$ may have two minima, but the lowest one is where the magnetization has the same sign as H . Therefore, if $H \neq 0$, the transition is not continuous. However, at $H = 0$ we do find a continuous transition. In particular for $H = 0$:

$$M_{eq} = \pm \sqrt{\frac{-A(T - T_c)}{B}}.$$

Note that given $A, B > 0$, the previous solutions have only physical meaning when $T < T_c$. For $T > T_c$ the only physical solution is $M_{eq} = 0$. The two degenerate minima are a consequence of the spin-flip symmetry occurring at $H = 0$. For $h \rightarrow 0^+$, the minimum with the $+$ sign wins, and likewise for the minus sign. Since the partition function is dominated by configurations with magnetization near M_{eq} , this means that the expectation value of the magnetization is

$$\langle M \rangle = M_0. \quad (12)$$

Problem 1: [2.5 points]

Consider a set of indistinguishable non-interacting particles ($i = 1, \dots, N$) moving in one dimension and subject to an external potential, described by the following classical Hamiltonian:

$$H = \sum_{i=1}^N \frac{p_i^2}{2m} + V(q_i),$$

where m is the mass of the particles, q_i are the position coordinates ($-\infty < q_i < +\infty$), p_i are the conjugate momenta, and the potential $V(q)$ is given by

$$V(q) = \begin{cases} \frac{1}{2}m\omega^2 q^2 + \varepsilon_0 & \text{if } q < 0, \\ \frac{1}{2}m\omega^2 q^2 & \text{if } q \geq 0, \end{cases}$$

where ε_0 is a positive constant. Find:

- (a) The canonical partition function $Q_N(T)$;
- (b) The internal energy $U(N, T)$;
- (c) The heat capacity.
- (d) Study the high and low-temperature limits and discuss them.

Solution Problem 1:

- (a) The single-particle partition function is given by

$$Q_1(T) = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dp dq e^{-\beta \left(\frac{p^2}{2m} + V(q) \right)}$$

where

$$Q_1(T) = \frac{1}{h} \int_{-\infty}^{\infty} dp e^{-\beta \frac{p^2}{2m}} \left[\int_{-\infty}^0 dq e^{-\beta \left(\frac{1}{2}m\omega^2 q^2 + \varepsilon_0 \right)} + \int_0^{\infty} dq e^{-\beta \frac{1}{2}m\omega^2 q^2} \right]$$

Simplifying further,

$$Q_1(T) = \frac{1}{h} \frac{\sqrt{2m\pi}}{\beta} (1 + e^{-\beta\varepsilon_0}) \frac{\sqrt{\pi}}{2} \frac{1}{\sqrt{\frac{1}{2}m\omega^2\beta}} = \frac{\pi}{\beta\hbar\omega} (1 + e^{-\beta\varepsilon_0}) = \frac{1}{\beta\hbar\omega} \frac{(1 + e^{-\beta\varepsilon_0})}{2}$$

And for N particles,

$$Q_N(T) = \frac{1}{N!} \left(\frac{1}{\beta\hbar\omega} \frac{(1 + e^{-\beta\varepsilon_0})}{2} \right)^N$$

- (b) The free energy is

$$\begin{aligned} F &= -k_B T \ln Q_N = -k_B T N \ln \left(\frac{1}{\beta\hbar\omega} \frac{(1 + e^{-\beta\varepsilon_0})}{2} \right) + k_B T N (\ln N - 1) \\ &= -k_B T N \left[\ln \left(\frac{1}{N\beta\hbar\omega} \frac{(1 + e^{-\beta\varepsilon_0})}{2} \right) + 1 \right] \end{aligned}$$

The entropy is given by

$$S = - \left(\frac{\partial F}{\partial T} \right)_{N,V} = k_B N \ln \left(\frac{1}{\beta \hbar \omega} \left(\frac{1 + e^{-\beta \epsilon_0}}{2} + 1 \right) \right) + k_B N + k_B N \frac{\frac{\epsilon_0}{k_B T} e^{-\beta \epsilon_0}}{e^{\beta \epsilon_0} + 1}$$

Then the internal energy

$$U = F + TS$$

is

$$U = k_B T N + N \epsilon_0 \frac{1}{e^{\beta \epsilon_0} + 1}$$

(c) Deriving respect to T we obtain the heat capacity:

$$C_v = k_B N + N k_B \left(\frac{\epsilon_0}{k_B T} \right)^2 \frac{e^{\beta \epsilon_0}}{(e^{\beta \epsilon_0} + 1)^2}$$

(d) For high and low temperatures we obtain:

$$C_v = k_B N$$

Problem 2:

Suppose a system of $s = 1/2$ non-interacting fermions which have a density of states $g(\epsilon) = A\epsilon$, $\epsilon \geq 0$.

- (a) Find the Fermi energy (i.e. the chemical potential at $T = 0$) for a system with N such particles.
- (b) Find the partition function assuming that $(\epsilon - \mu) \gg k_B T$.
- (c) Find the average energy $\langle E \rangle$ in the above limit and relate it to PV .

NOTE: If needed use the definition $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$.

Solution Problem 2:

- (a) Find the Fermi energy (i.e. the chemical potential at $T = 0$) for a system with N such particles.

At $T = 0$ only energy levels with $\epsilon \leq \mu(T = 0)$ are occupied with probability 1 so that $f(\epsilon) = 1$ for $\epsilon < \mu = \epsilon_F$ and zero otherwise. By setting the expected number of particles to N we get that

$$N = \int_0^\infty d\epsilon g(\epsilon) f(\epsilon) = \int_0^\infty d\epsilon g(\epsilon) = \frac{A}{2} \epsilon_F^2 \quad \epsilon_F = \sqrt{\frac{2N}{A}}. \quad (13)$$

NOTE: I am assuming that the spin degeneracy is already taken into account in A . However, if you considered it is not then you would have to multiply results by an extra factor of 2.

- (b) Find the partition function at assuming that $(\epsilon - \mu) \gg k_B T$.

As we have seen in class, the partition function is

$$\mathcal{Z} = \prod_i \left(1 + e^{-\beta(\epsilon_i - \mu)}\right) \quad (14)$$

which in the limit $\beta(\epsilon - \mu) \gg 1$ goes to 1.

- (c) Find the average energy $\langle E \rangle$ at high-temperature and relate it to PV .

We have that we can obtain PV from the logarithm of the partition function as:

$$\begin{aligned} PV &= k_B T \log \mathcal{Z} = k_B T \sum_i \log \left(1 + e^{-\beta(\epsilon_i - \mu)}\right) \\ &\approx k_B T \int_0^\infty d\epsilon g(\epsilon) \log \left(1 + e^{-\beta(\epsilon - \mu)}\right) \\ \lim_{\beta(\epsilon - \mu) \gg 1} &\rightarrow \int_0^\infty d\epsilon g(\epsilon) e^{-\beta(\epsilon - \mu)} = k_B T \int_0^\infty d\epsilon g(\epsilon) e^{-\beta(\epsilon - \mu)} \\ &= \int_0^\infty d\epsilon A \epsilon e^{-\beta(\epsilon - \mu)} = (k_B T)^3 A e^{\beta\mu} \int_0^\infty x e^{-x} \\ &= (k_B T)^3 A e^{\beta\mu} \Gamma(2), \end{aligned}$$

which yields the following equation of state:

$$PV = (k_B T)^3 A e^{\beta\mu} \Gamma(2) \quad (15)$$

where we have used that $\log(1 + x) \approx x$ for $x \ll 1$ and in the last line, we have applied the change of variable $x = \beta\epsilon$.

For the energy, we apply a similar procedure or we take the partial derivative:

$$\langle E \rangle = - \left(\frac{\partial \log \mathcal{Z}}{\partial \beta} \right)_{\beta\mu} \quad (16)$$

$$= - \left(\frac{\partial}{\partial \beta} \beta^{-2} A e^{\beta\mu} \Gamma(2) \right)_{\beta\mu} = 2(k_B T)^3 A e^{\beta\mu} \Gamma(2) \quad (17)$$

so that

$$PV = \frac{\langle E \rangle}{2} \quad (18)$$

in the limit $\beta(\epsilon - \mu) \gg 1$.